GLOBAL HEALTH CARE STATISTICS REPORT

ICA Group Project

The dataset contains 80,277 records and 22 attributes on global health statistics: disease prevalence, incidence and mortality, healthcare access, demographics, treatment types and socioeconomic factors. It covers 20 countries and 20 diseases over two years (2023 and 2024). The analysis was carried out in Python using pandas, NumPy, SciPy, Matplotlib and Seaborn.

A. System Study / Domain Analysis (5 Marks)

ANS 1. Business Process: Global Health Monitoring and Policymaking

The dataset supports the assessment and improvement of public health outcomes across regions, age groups and genders. It supports four activities:

- **Tracking disease trends.** Prevalence shows the share of a population affected, incidence shows new cases over a period, and mortality shows which diseases are deadliest. If a disease shows a rising trend in a region, governments can focus on prevention there, for example through awareness campaigns.
- **Assessing healthcare access.** Healthcare access (%), doctors per 1,000 and hospital beds per 1,000 show whether services are adequate. A region with few doctors and beds but high infectious disease incidence may need urgent healthcare expansion.
- **Planning resource allocation.** Demographic data (age group, gender, population affected) combined with income and education indicators lets governments and organisations direct funding, supplies and staff to where need is highest.
- **Evaluating interventions.** Recovery rate, DALYs (Disability-Adjusted Life Years) and the 5-year improvement figure show whether treatments and campaigns are working, for example whether recovery rates improved after a vaccination drive.

Types of analytics

- **Descriptive:** summarises prevalence, incidence and mortality by region, age group and gender, and relates healthcare access to outcomes. A region with high mortality and low healthcare access points to a systemic problem.
- **Predictive:** uses historical data to forecast disease trends and resource demand from factors such as urbanisation, income and education. For example, forecasting a winter rise in respiratory disease to plan vaccine supply.
- **Prescriptive:** recommends where to place hospital beds, funding or staff, and which policies to adopt. If DALYs stay high despite treatment availability, policy may need to focus on access or cost.

Usefulness for the business process

- **Public health policy:** prioritise high-prevalence and high-mortality diseases and respond early to outbreaks. Impact: lower disease burden.
- **Healthcare planning:** allocate vaccines, beds and personnel equitably. Impact: more efficient health systems.

- **Economic impact:** treatment cost, DALYs and income data show the financial burden of disease and support the case for funding. Impact: less financial strain on systems and individuals.
- **Targeted interventions:** segmenting by age, gender and geography allows precise targeting, for example vaccination campaigns in groups with low recovery rates. Impact: better outcomes for at-risk groups.

ANS 2. Data Composition

The table lists each attribute, its type, the number of missing values in the raw data, and its role in the analysis.

Attribute	Type	Missing	Role
Country	Categorical	13	Geographic grouping
Year	Numerical (discrete)	25	Time period
Disease Name	Categorical	15	Disease identifier
Disease Category	Categorical	20	Disease grouping
Prevalence Rate (%)	Numerical	20	Disease burden
Incidence Rate (%)	Numerical	20	Disease burden
Mortality Rate (%)	Numerical	12	Outcome
Age Group	Categorical	20	Demographic
Gender	Categorical	20	Demographic
Population Affected	Numerical (discrete)	79	Scale of disease burden
Healthcare Access (%)	Numerical	39	Healthcare resources
Doctors per 1000	Numerical	39	Healthcare resources
Hospital Beds per 1000	Numerical	17	Healthcare resources
Treatment Type	Categorical	17	Treatment
Average Treatment Cost (USD)	Numerical (discrete)	47	Treatment cost
Availability of Vaccines/Treatment	Categorical	35	Treatment availability
Recovery Rate (%)	Numerical	35	Outcome
DALYs	Numerical (discrete)	35	Outcome
Improvement in 5 Years (%)	Numerical	47	Outcome
Per Capita Income (USD)	Numerical (discrete)	47	Socioeconomic
Education Index	Numerical	47	Socioeconomic
Urbanization Rate (%)	Numerical	47	Socioeconomic

Note: an earlier version of this report rated each attribute's importance as High, Average or Low. Those ratings were assumptions based on domain knowledge, not measurements. They are replaced here with each attribute's role, and feature relevance is tested with correlations in Section C.

B. Data Cleaning (5 Marks)

ANS 1. Analysis of Data Types

When loaded, pandas reads all numerical columns as float64 (because they contain missing values) and text columns as strings. Most columns are correctly typed. Five columns hold whole numbers and were converted to integers:

Column	Original type	Converted to	Reason
Year	float64	int64	Calendar year
Population Affected	float64	int64	Count of people
Average Treatment Cost (USD)	float64	int64	Whole dollars
DALYs	float64	int64	Whole years
Per Capita Income (USD)	float64	int64	Whole dollars

The conversion is done after missing values are filled, because integer columns cannot hold missing values.

ANS 2. Summary and Handling of Missing Values

Missing values were counted with df.isnull().sum() (the per-column counts are in the table above). In total there are 696 missing cells out of 1,766,094 (0.04%). The most affected column is Population Affected, with 79 missing values (0.10% of rows). Because so little is missing, imputation is preferable to deleting rows, and its effect on the results is minimal.

Column group	Method	Reason
Categorical (Country, Disease Name, Disease Category, Age Group, Gender, Treatment Type, Vaccine Availability) and Year	Mode	Most frequent value is the natural fill for labels
Rates and indices (prevalence, incidence, mortality, recovery, healthcare access, doctors, beds, 5-year improvement, education, urbanisation)	Mean	Distributions are symmetric, so the mean is representative
Counts, costs and income (Population Affected, Treatment Cost, DALYs, Per Capita Income)	Median	Robust to extreme values

Data cleaning results

- **Missing values:** 0 remaining after imputation (from 696).
- **Data types:** Year, Population Affected, Average Treatment Cost, DALYs and Per Capita Income converted to int64.
- **Duplicates:** 0 duplicate rows found.
- **Final shape:** 80,277 rows x 22 columns.

C. Exploratory Data Analysis (10 Marks)

ANS 1. Univariate Analysis

a. Numerical variables

Summary statistics for the main numerical variables (after cleaning):

Variable	Mean	Std dev	Min	Median	Max
Prevalence Rate (%)	10.03	5.74	0.10	10.02	20.00
Incidence Rate (%)	7.55	4.29	0.10	7.56	15.00
Mortality Rate (%)	5.05	2.86	0.10	5.04	10.00
Healthcare Access (%)	75.00	14.46	50.00	75.07	100
Recovery Rate (%)	74.54	14.18	50.00	74.53	99.00
DALYs	2,500	1,444	1.00	2,499	5,000
Per Capita Income (USD)	50,333	28,726	502	50,380	100,000
Education Index	0.65	0.14	0.40	0.65	0.90

Boxplots. A boxplot shows the median, the interquartile range (IQR, the middle 50% of values) and outliers (points beyond 1.5 x IQR). DALYs range from 1 to 5,000 with a median of 2,499 (Q1 = 1,247, Q3 = 3,752). Per Capita Income ranges from USD 502 to USD 100,000 with a median of USD 50,380 (Q1 = 25,451, Q3 = 75,178). In both cases the whiskers extend to the minimum and maximum, so there are no outliers.

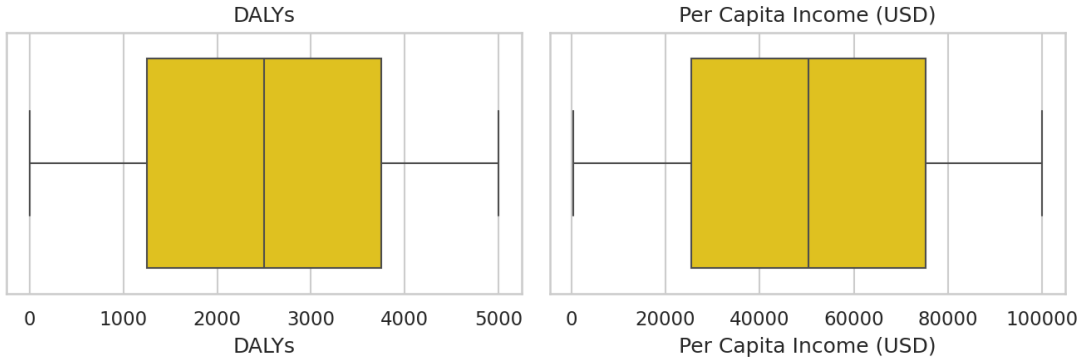


Figure 1: Boxplots of DALYs and Per Capita Income (USD)

Distributions and normality. Skewness measures asymmetry: a right-skewed variable has a long tail of high values, a left-skewed one a long tail of low values. The scipy normaltest function tests the null hypothesis (H0) that data is normally distributed; a p-value above 0.05 means we fail to reject H0.

Variable	Skewness	normaltest p-value	Normal?
Prevalence Rate (%)	0.0095	< 0.001	No
Mortality Rate (%)	0.0018	< 0.001	No
Healthcare Access (%)	-0.0082	< 0.001	No
DALYs	0.0035	< 0.001	No
Per Capita Income (USD)	-0.0061	< 0.001	No

All variables have skewness very close to zero, so they are symmetric. However, every p-value is below 0.001, so none is normally distributed. The histograms show why: the values are spread almost evenly across their range (a flat, uniform-like shape) instead of forming a bell curve. With 80,000 records the test also detects even tiny deviations, so the histograms are more informative than the p-value alone. Symmetry here comes from the flat shape of the data, not from the large sample size.

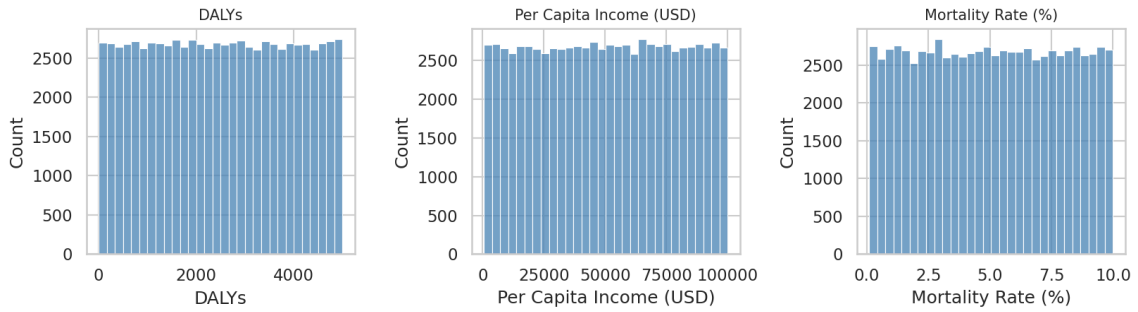


Figure 2: Distributions of DALYs, Per Capita Income and Mortality Rate

b. Categorical variables

Bar charts show how many records fall in each category. The number of records per disease ranges from 3,893 (Tuberculosis) to 4,116 (Cholera), a gap of only 5.6% of the average. Cholera has the most records and Tuberculosis the fewest, but the difference is small, so the diseases are almost evenly represented rather than one being clearly "most common".

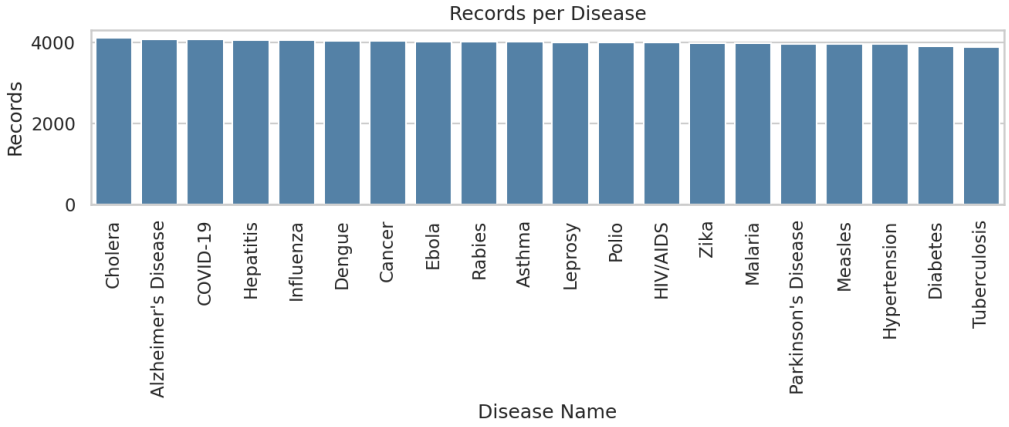


Figure 3: Number of records per disease

The pie chart of disease categories shows the same balance: shares range from 8.91% (Metabolic) to 9.3% (Bacterial). It is therefore not possible to conclude that any single type of disease, such as viral infections, dominates the dataset.

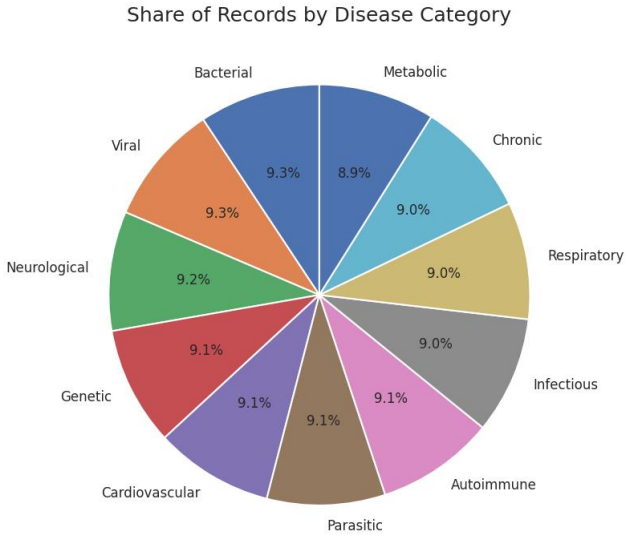


Figure 4: Share of records by disease category

Age group, gender and treatment type are also close to evenly split (age groups range from 19,946 to 20,210 records). Because these are categories, bar charts are used instead of line graphs, which imply an order or trend that does not exist between categories. The small difference between age groups (about 1.3% of the average) does not show that older people are more likely to be ill.

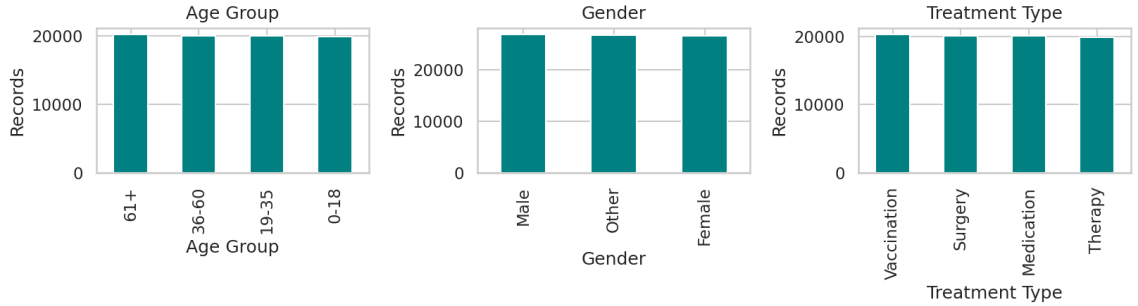


Figure 5: Records by age group, gender and treatment type

ANS 2. Bivariate Analysis

a. Relationships between variables

Time series. The data covers only two years, so a trend is a change between two points. Average mortality rate moved from 5.044% in 2023 to 5.050% in 2024, a change of 0.13%. The other indicators changed by less than 0.3%. These changes are too small to be called a rise or fall.

Variable (mean)	2023	2024	Change
Mortality Rate (%)	5.044	5.050	0.13%
Prevalence Rate (%)	10.048	10.019	-0.29%
Healthcare Access (%)	75.031	74.978	-0.07%
Per Capita Income (USD)	50,294	50,372	0.15%
Education Index	0.651	0.651	-0.04%

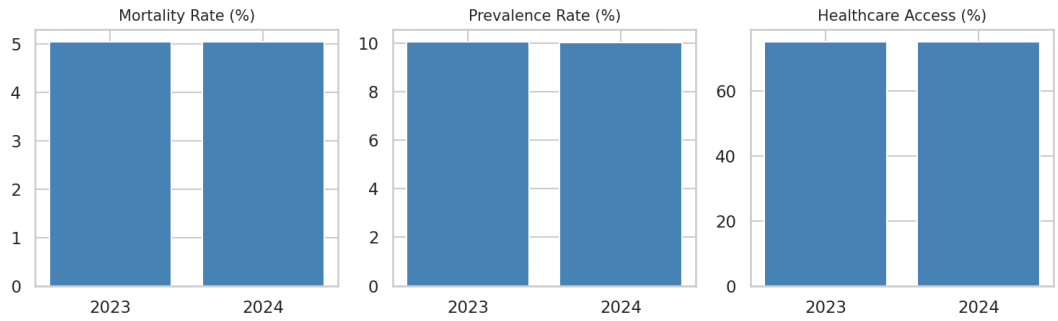


Figure 6: Average mortality rate, prevalence rate and healthcare access, 2023 vs 2024 (y-axis starts at zero)

Pivot table. A pivot table summarises variables by country and year. Mortality and recovery are rates (percentages), so they are averaged; adding percentages together would not be meaningful. Population Affected is a count, so it is summed. The first rows are shown below.

Country	Year	Mean mortality (%)	Mean recovery (%)	Population affected
Argentina	2023	5.09	74.47	966,934,791
Argentina	2024	5.05	74.45	991,206,406
Australia	2023	4.94	74.52	1,019,013,359
Australia	2024	5.02	74.47	995,795,256
Brazil	2023	5.06	74.77	988,892,497
Brazil	2024	5.15	74.13	990,232,279
Canada	2023	5.09	74.61	1,025,801,015
Canada	2024	5.11	74.71	976,222,967

Across countries, average mortality ranges only from 4.93% (USA) to 5.11% (South Korea), so no country stands out.

Correlation and scatter plots. The correlation matrix measures the linear relationship (r, from -1 to +1) between each pair of numerical variables. The strongest correlation between any two variables is $|r| = 0.009$, which is effectively zero.

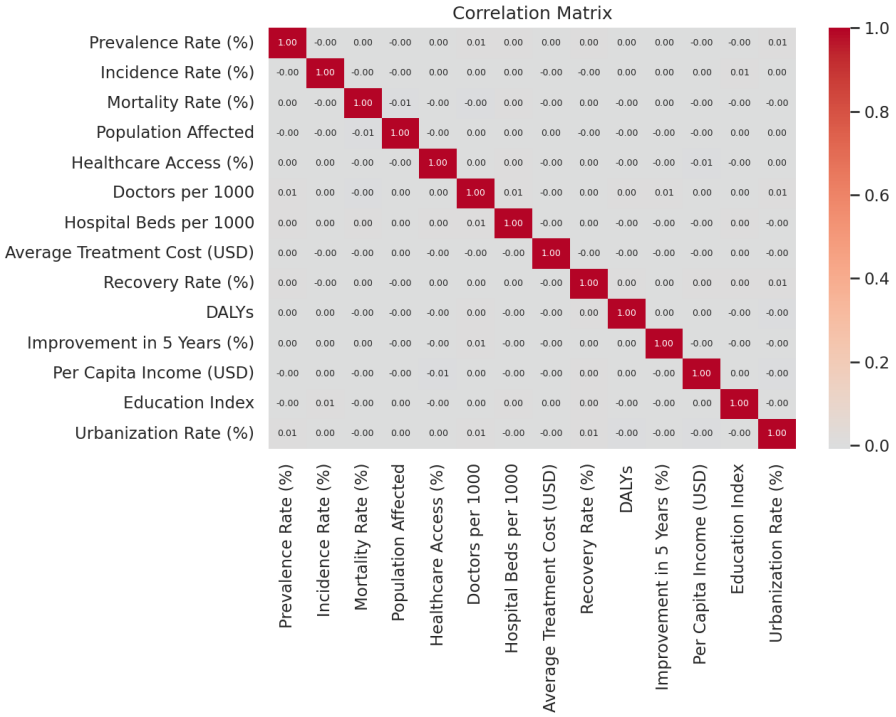


Figure 7: Correlation matrix of numerical variables

Outcome	Variable	Correlation (r)
Mortality Rate (%)	Population Affected	-0.008
Mortality Rate (%)	Doctors per 1000	-0.005
Mortality Rate (%)	Hospital Beds per 1000	0.004
Recovery Rate (%)	Urbanization Rate (%)	0.006
Recovery Rate (%)	DALYs	0.004
Recovery Rate (%)	Per Capita Income (USD)	0.004

The scatter plots below use the full range of each axis, a random sample of 5,000 points for readability, and a fitted regression line. In each case the cloud of points is shapeless and the line is flat.

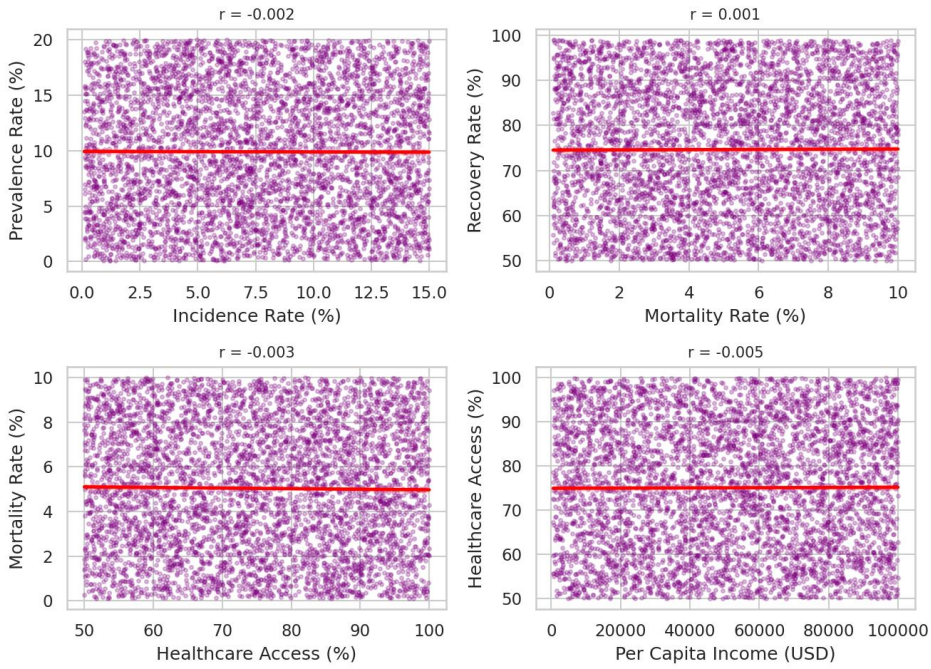


Figure 8: Scatter plots with regression lines (r computed on all records)

Vaccine and treatment availability. Mean mortality is 5.044% where vaccines/treatment are not available and 5.049% where they are. Mean recovery is 74.56% and 74.52% respectively. The difference is negligible.

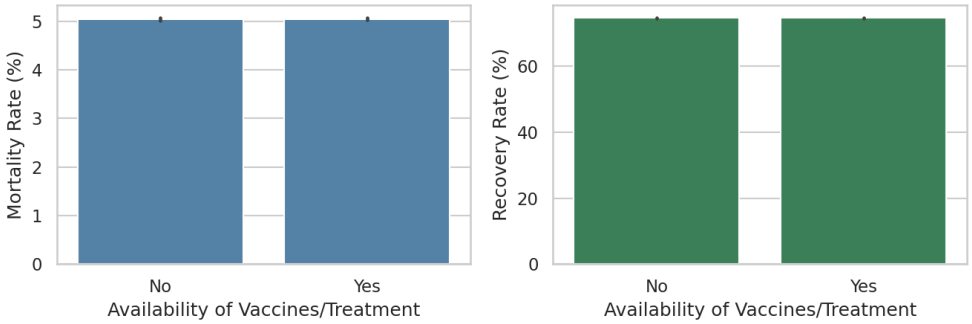


Figure 9: Mean mortality and recovery rate by vaccine/treatment availability

b. Feature importance from bivariate analysis

Domain knowledge suggests that healthcare access, vaccine availability, doctors per 1,000, income and education should influence mortality and recovery. The data does not support this. No feature has a correlation above 0.01 with mortality or recovery, and vaccine availability makes no visible difference. This means the bivariate analysis cannot rank any feature as important for predicting these outcomes. If a predictive model were built on this data, we would expect it to perform no better than predicting the average.

Conclusions and Limitations

- **Data quality:** the dataset is clean and complete after imputing 696 missing cells (under 0.04%), with no duplicates.
- **Balance:** diseases, disease categories, age groups, genders and treatment types are all almost evenly represented.

- **Relationships:** no meaningful relationship exists between healthcare resources, socioeconomic indicators and health outcomes (all $|r| < 0.01$).
- **Likely synthetic data:** the flat distributions, the near-equal category counts and the absence of any correlation strongly suggest that the values were generated randomly. In real health data, income, healthcare access and mortality would normally be clearly related. The results should therefore be read as a demonstration of data-cleaning and EDA method, not as findings about actual global health.
- **Other limitations:** only two years of data, so no long-term trends can be shown; imputation with mean, median and mode slightly reduces variance; and correlation does not imply causation.